This assignment will not be graded and is only for practice.

2.1 Level 1:

1) Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -2 & 9 & -6 \\ -3 & 4 & 7 & 7 \end{bmatrix}$. Which of the following is true about the matrix A? **1 point**

[Hint: The (i, j)-th entry is the entry which is at the i-th row and j-th column.]

- [ ] It is a 4×3 matrix.
- [x] It is a 3×4 matrix.
- [ ] $(2, 3)$-th entry of the matrix A is 4.
- [x] $(2, 3)$-th entry of the matrix A is 9.

Yes, the answer is correct.

Score: 1

Accepted Answers:

It is a 3×4 matrix.
$(2, 3)$-th entry of the matrix A is 9.

2) Which of the following statements is(are) TRUE? **1 point**

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

- ( ) Any diagonal matrix is a scalar matrix.
- ( ) Scalar matrices may not be square matrices.
- (•) Scalar matrices must be square matrices.
- ( ) Any scalar matrix is an identity matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Scalar matrices must be square matrices.

3) **1 point**

Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 36$$

$$8x_1 + 4x_2 - 9x_3 = 11$$

$$4x_1 - x_2 + 3x_3 = 10$$

If the matrix representation of the system of equations is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

- [x] $A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- [ ] $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & 9 & 11 \\ 4 & 1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- [ ] $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & -9 & 11 \\ 4 & -1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- [ ] $A = \begin{bmatrix} 7 & 10 & 36 & 12 \\ 8 & 4 & 11 & -9 \\ 4 & -1 & 10 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$

4) Which of the following statements is(are) TRUE? ***1 point***

- [ ] Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.
- [x] Addition of two matrices is possible only if the orders of both the matrices are the same.
- [ ] Defining AB is possible if the number of rows in the matrix A is same as the number of columns in the matrix B.

- [x] Defining AB is possible if the number of columns in the matrix A is same as the number of rows in the matrix B.

Yes, the answer is correct.
Score: 1

Accepted Answers:

Addition of two matrices is possible only if the orders of both the matrices are the same.
Defining AB is possible if the number of columns in the matrix A is same as the number of rows in the matrix B.

5) Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$, $Q = [1 \quad 4 \quad -9]$, $R = [0 \quad -3 \quad 10]$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$ ***1 point***

[Hint: If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$, then the order of AB is $m \times p$.]

- [x] The matrix PD is of order 3×1.
- [ ] The matrix PD is of order 1×3.
- [ ] The matrix QD is of order 3×3.
- [x] The matrix QD is of order 1×1.
- [x] The matrix DQ is of order 3×3.
- [ ] The matrix DQ is of order 1×1.
- [ ] The product QD is not defined.
- [x] The product QR is not defined.
- [x] The addition $P + Q$ is not defined.
- [x] The addition $P + D$ is not defined.

Yes, the answer is correct.
Score: 1

Accepted Answers:

The matrix PD is of order 3×1.

The matrix QD is of order 1×1.
The matrix DQ is of order 3×3.
The product QR is not defined.
The addition $P + Q$ is not defined.
The addition $P + D$ is not defined.

2.2 Level 2:

6) Choose the set of correct options. ***1 point***

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given conditions to find matrix A, for first two options.]

- [ ] If A is a square matrix of order 2 and $A^2 = I$, then $A = I$ or $A = -I$ where I is the identity matrix of order 2.
- [ ] If A is a square matrix of order 2 and $A^2 = 0$, then $A = 0$.
- [x] If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.
- [x] If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

Yes, the answer is correct.
Score: 1

Accepted Answers:

If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.
If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

7) If A is a square matrix of order 2 whose first column is denoted by C_1 and second column is denoted by C_2 (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ then choose the set of correct options. ***1 point***
[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]

- [x] The first column of AB is $b_{11}C_1 + b_{21}C_2$.
- [x] The second column of AB is $b_{12}C_1 + b_{22}C_2$.
- [ ] The first column of AB is $b_{21}C_1 + b_{11}C_2$.
- [ ] The second column of AB is $b_{22}C_1 + b_{21}C_2$.

Yes, the answer is correct.
Score: 1

Accepted Answers:

The first column of AB is $b_{11}C_1 + b_{21}C_2$.
The second column of AB is $b_{12}C_1 + b_{22}C_2$.

8) Choose the set of correct options. **1 point**

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given condition to find matrix A for last two options.]

- [x] There exist some real matrices A and B, such that $AB = BA$.
- [ ] There do not exist any real matrices A and B, such that $AB = BA$.
- [ ] There does not exist any real matrix A, such that $A^2 = A$.
- [x] There exists some real 2×2 matrix A, such that $A^2 + A + I = 0$

Yes, the answer is correct.

Score: 1

Accepted Answers:

There exist some real matrices A and B, such that $AB = BA$.
There exists some real 2×2 matrix A, such that $A^2 + A + I = 0$

9) Suppose A is a 3×3 scalar matrix and (1,1)-th entry of the matrix A is 4. Suppose B is a 3 $\times$ 3 square matrix such that (i, j)-th entry is equal to $i^2 + j^2$. Find the (2,2)-th entry of the matrix $2A + B$.

16

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

1 point

10) Suppose $A = \begin{bmatrix} -1 & 2 \\ -4 & 7 \end{bmatrix}$ and $A^2 - \alpha A + I = 0$ for some $\alpha \in \mathbb{R}$.

Find the value of α.

6

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 6

1 point

Your score is: 10/10